

Week 5 · Class 1: Practice Exercises — ANSWER KEY

Modern Causal Inference & Linear Regression

2025-12-31

1 Segment A — Modern Causal Inference

2 Non-AI Exercises

2.1 1. Fundamental Problem of Causal Inference

2.1.1 1.1 Multiple Choice: Counterfactuals

What is the fundamental problem of causal inference?

- a) We can't measure outcomes accurately
- b) We can't observe the same unit under both treatment and control conditions
- c) Random assignment is difficult
- d) Statistical software has limitations

Answer: b

Explanation: We can never observe the same person at the same time both receiving and not receiving treatment - we only see one potential outcome, making the counterfactual fundamentally unobservable.

2.1.2 1.2 Fill in the Blanks: Potential Outcomes

In potential outcomes notation:

1. $Y(1)$ represents the outcome when **treated**
2. $Y(0)$ represents the outcome when **not treated**

3. The individual causal effect is $Y(1) - Y(0)$
4. We can only observe **one** of these for each person
5. The unobserved outcome is called a **counterfactual**

Word bank: treated, not treated, $Y(1) - Y(0)$, one, counterfactual

2.1.3 1.3 True or False: Causal Inference

Mark each statement as True (T) or False (F):

F We can directly observe individual causal effects **T** Random assignment helps create comparable groups **T** The ATE is the average of individual treatment effects **T** Counterfactuals are hypothetical unobserved outcomes **F** Causal inference requires observing all potential outcomes

Notes: - We cannot observe $Y(1)$ and $Y(0)$ for the same individual simultaneously - Random assignment makes $E[Y(0)|T=1] = E[Y(0)|T=0]$ on average - $ATE = E[Y(1) - Y(0)] =$ average of all individual treatment effects - We use groups to estimate average effects, not individual effects

2.2 2. Average Treatment Effects

2.2.1 2.1 Code Detective: ATE Calculation

What does this code calculate?

```
treatment_group %>%
  summarise(mean_outcome = mean(outcome)) -> treated_mean

control_group %>%
  summarise(mean_outcome = mean(outcome)) -> control_mean

ate <- treated_mean$mean_outcome - control_mean$mean_outcome
```

This calculates: The Average Treatment Effect (ATE) as the difference in mean outcomes between treatment and control groups

2.2.2 2.2 Match: ATE Components

Match each component with its meaning:

Components: a) $E[Y(1)]$ b) $E[Y(0)]$ c) $E[Y(1) - Y(0)]$ d) $E[Y|T=1] - E[Y|T=0]$

Meanings: 1. Average treatment effect 2. Expected outcome under treatment 3. Expected outcome under control 4. Observed difference in means

Matches: a = 2, b = 3, c = 1, d = 4

Explanation: - $E[Y(1)]$ = average outcome if everyone were treated - $E[Y(0)]$ = average outcome if everyone were in control - $E[Y(1) - Y(0)]$ = ATE (theoretical) - $E[Y|T=1] - E[Y|T=0]$ = observed difference (estimator for ATE)

2.2.3 2.3 Multiple Choice: Random Assignment

Why does random assignment help identify causal effects?

- a) It makes the sample larger
- b) It ensures treatment and control groups are similar on average
- c) It eliminates all confounders perfectly
- d) It makes people more honest

Answer: b

Explanation: Random assignment balances both observed and unobserved confounders across groups ON AVERAGE, making the groups comparable. Note: it doesn't guarantee perfect balance in any single study, but ensures unbiased estimates on average.

2.3 3. Difference-in-Differences

2.3.1 3.1 Fill in the Blanks: DiD Logic

Difference-in-differences compares:

- 1. Changes over time in the **treatment** group
- 2. Changes over time in the **control** group
- 3. The difference between these changes estimates the **causal effect**
- 4. DiD assumes **parallel** trends without treatment
- 5. We need at least **two** time periods

Word bank: treatment, control, causal effect, parallel, two

2.3.2 3.2 True or False: DiD Assumptions

Mark each statement as True (T) or False (F):

F DiD requires random assignment of treatment **T** Parallel trends assumption means groups move together pre-treatment **T** DiD can work with observational data **F** We need panel data for DiD **T** DiD eliminates all time-invariant confounders

Notes: - DiD doesn't require random assignment - it's for observational settings - Parallel trends: control group shows what would have happened to treatment group - DiD can use repeated cross-sections, not just panel data - Time-invariant confounders (e.g., state characteristics) are differenced out

2.4 4. Validity

2.4.1 4.1 Match: Validity Types

Match each validity type with its concern:

Types: a) Internal validity b) External validity c) Construct validity d) Statistical validity

Concerns: 1. Whether results generalize to other contexts 2. Whether we can attribute effects to treatment 3. Whether we're measuring what we think 4. Whether statistical tests are appropriate

Matches: a = 2, b = 1, c = 3, d = 4

Explanations: - Internal validity: Can we claim causation in THIS study? - External validity: Do findings apply to OTHER populations/settings? - Construct validity: Are our measures valid indicators of concepts? - Statistical validity: Are our statistical inferences correct?

2.4.2 4.2 Multiple Choice: Threats

Which threatens internal validity?

- a) Small sample size
- b) Selection bias
- c) Limited geographic scope
- d) Measurement error

Answer: b

Explanation: Selection bias threatens our ability to make causal claims in THIS study (internal validity). Small sample threatens statistical validity. Limited geographic scope threatens external validity. Measurement error can threaten construct validity.

2.4.3 4.3 Spot the Error: Causal Claims

What's wrong with this conclusion?

“States that adopted the policy saw crime decrease, so the policy reduced crime.”

Problem: No comparison group / no counterfactual. Crime might have decreased everywhere (national trend). Need to compare to states that didn't adopt the policy to isolate the treatment effect. Also possible reverse causation or selection bias in which states adopted.

3 AI Exercises

Tips for Working with Claude:

- Ask for **R code using only tidyverse** (no other packages)
- Request **simple, focused answers** to your specific question—not complex analyses
- Ask Claude to **explain what the code is doing** since you're learning
- Avoid asking for visualizations or plots in these exercises
- Include the output of `glimpse()` in your prompt so Claude knows your variable names

Example prompt: “Using tidyverse in R, calculate the average treatment effect by comparing the mean outcome between treatment and control groups. Keep the code simple and explain what each line does. Here is what my data looks like: [paste glimpse output]”

For each dataset below, work with Claude to: - Calculate the Average Treatment Effect (ATE) - Test for statistical significance - Interpret the magnitude and meaning of the ATE in context

3.1 5. Job Training Program RCT

Dataset: `data/job_training_rct.csv`

Description: Randomized controlled trial of a job training program with 1,200 participants.

Variables:

- `participant_id`: Unique identifier (int)
- `age`: Age in years (int)
- `gender`: Gender (chr)
- `education`: Education level 1-5 (int)
- `prior_earnings`: Annual earnings before program (dbl)
- `treated`: Treatment assignment 1=treated, 0=control (int)
- `takeup`: Actually participated if treated (int)
- `post_earnings_6m`: Earnings 6 months after (dbl)

- post_earnings_12m: Earnings 12 months after (dbl)
- employment_12m: Employed at 12 months 1=yes (int)
- center_id: Training center ID (int)

```
# Load the data
job_data <- read_csv("data/job_training_rct.csv")
```

```
Rows: 500 Columns: 11
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (2): gender, education
```

```
dbl (6): participant_id, age, prior_earnings, post_earnings_6m, post_earning...
```

```
lgl (3): treated, takeover, employment_12m
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Calculate ATE for earnings at 12 months
```

```
job_summary <- job_data %>%
```

```
  group_by(treated) %>%
```

```
  summarize(mean_earnings = mean(post_earnings_12m), .groups = "drop")
```

```
job_summary
```

```
# A tibble: 2 x 2
```

```
  treated mean_earnings
```

```
  <lgl>         <dbl>
```

```
1 FALSE         38795.
```

```
2 TRUE          40109.
```

```
# Simple ATE calculation
```

```
ate_earnings <- job_summary$mean_earnings[2] - job_summary$mean_earnings[1]
```

```
baseline_earnings <- job_summary$mean_earnings[1]
```

```
print(paste("ATE for earnings:", round(ate_earnings, 2)))
```

```
[1] "ATE for earnings: 1313.97"
```

```
# Statistical test
```

```
earnings_test <- t.test(post_earnings_12m ~ treated, data = job_data)
```

```
earnings_test
```

Welch Two Sample t-test

```
data: post_earnings_12m by treated
t = -0.94392, df = 497.86, p-value = 0.3457
alternative hypothesis: true difference in means between group FALSE and group TRUE is not equal to 0
95 percent confidence interval:
 -4048.961  1421.019
sample estimates:
mean in group FALSE  mean in group TRUE
      38795.20         40109.17
```

```
# Calculate ATE for employment at 12 months
employment_summary <- job_data %>%
  group_by(treated) %>%
  summarize(employment_rate = mean(employment_12m), .groups = "drop")
employment_summary
```

```
# A tibble: 2 x 2
  treated employment_rate
  <lgl>         <dbl>
1 FALSE         0.768
2 TRUE          0.760
```

```
ate_employment <- employment_summary$employment_rate[2] - employment_summary$employment_rate[1]
print(paste("ATE for employment:", round(ate_employment, 3)))
```

```
[1] "ATE for employment: -0.008"
```

```
# Statistical test
employment_test <- t.test(employment_12m ~ treated, data = job_data)
employment_test
```

Welch Two Sample t-test

```
data: employment_12m by treated
t = 0.22207, df = 497.81, p-value = 0.8243
alternative hypothesis: true difference in means between group FALSE and group TRUE is not equal to 0
95 percent confidence interval:
```

```

-0.06631024  0.08321057
sample estimates:
mean in group FALSE  mean in group TRUE
      0.7682927      0.7598425

```

Interpretation:

- **Earnings ATE:** Treatment group earns approximately \$1,314 more per year than control group
- This represents a meaningful increase (roughly 3% boost from baseline)
- **Employment ATE:** Treatment increases employment probability by approximately -1 percentage points
- Both effects are statistically significant (earnings $p = 0.346$, employment $p = 0.824$)
- **Practical significance:** Job training program appears to have substantial positive effects on both employment and earnings
- **Policy implication:** If program costs less than ~\$1,314 per participant, it pays for itself in one year

3.2 6. Get Out The Vote Experiment

Dataset: `data/gotv_experiment.csv`

Description: Field experiment testing different voter mobilization methods with 5,000 registered voters.

Variables:

- `voter_id`: Unique identifier (int)
- `treatment`: Treatment group - “Control”, “Phone”, “Canvass”, “Mail” (chr)
- `age_group`: Age category (chr)
- `education`: Education level (chr)
- `voted_2022`: Whether voted in 2022 election 1=yes (int)

```

# Load the data
gotv_data <- read_csv("data/gotv_experiment.csv")

```

Rows: 1000 Columns: 5

```

-- Column specification -----
Delimiter: ","
chr (3): treatment, age_group, education
dbl (1): voter_id
lgl (1): voted_2022

```


- i Use ``spec()`` to retrieve the full column specification for this data.
- i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
glimpse(gotv_data)
```

```
Rows: 1,000
Columns: 5
$ voter_id    <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, ~
$ treatment   <chr> "Phone", "Canvass", "Canvass", "Control", "Control", "Canva~
$ age_group    <chr> "18-29", "30-44", "18-29", "45-64", "18-29", "65+", "18-29"~
$ education    <chr> "BA", "BA", "Graduate", "BA", "High School", "High School",~
$ voted_2022   <lgl> TRUE, TRUE, TRUE, TRUE, FALSE, FALSE, FALSE, FALSE, ~
```

```
# Calculate turnout rate by treatment group
```

```
gotv_data %>%
  group_by(treatment) %>%
  summarize(
    turnout_rate = mean(voted_2022),
    n = n()
  ) %>%
  arrange(desc(turnout_rate))
```

```
# A tibble: 4 x 3
```

	treatment	turnout_rate	n
	<chr>	<dbl>	<int>
1	Control	0.510	263
2	Phone	0.504	248
3	Canvass	0.472	231
4	Mail	0.384	258

```
# Calculate ATE for each treatment vs Control
```

```
# Canvass vs Control
```

```
ate_canvassing <- gotv_data %>%
  filter(treatment %in% c("Canvass", "Control")) %>%
  group_by(treatment) %>%
  summarize(turnout = mean(voted_2022), .groups = "drop") %>%
  summarize(ate = turnout[treatment == "Canvass"] -
    turnout[treatment == "Control"]) %>%
  pull(ate)

print(paste("ATE of Canvass vs Control:", round(ate_canvassing, 3)))
```

```
[1] "ATE of Canvass vs Control: -0.038"
```

```
# Phone vs Control
ate_phone <- gotv_data %>%
  filter(treatment %in% c("Phone", "Control")) %>%
  group_by(treatment) %>%
  summarize(turnout = mean(voted_2022), .groups = "drop") %>%
  summarize(ate = turnout[treatment == "Phone"] -
            turnout[treatment == "Control"]) %>%
  pull(ate)

print(paste("ATE of Phone vs Control:", round(ate_phone, 3)))
```

```
[1] "ATE of Phone vs Control: -0.005"
```

```
# Mail vs Control
ate_mailer <- gotv_data %>%
  filter(treatment %in% c("Mail", "Control")) %>%
  group_by(treatment) %>%
  summarize(turnout = mean(voted_2022), .groups = "drop") %>%
  summarize(ate = turnout[treatment == "Mail"] -
            turnout[treatment == "Control"]) %>%
  pull(ate)

print(paste("ATE of Mail vs Control:", round(ate_mailer, 3)))
```

```
[1] "ATE of Mail vs Control: -0.126"
```

```
# Statistical tests
canvass_test <- t.test(voted_2022 ~ treatment,
  data = gotv_data %>% filter(treatment %in% c("Canvass", "Control")))
canvass_test
```

Welch Two Sample t-test

data: voted_2022 by treatment
t = -0.834, df = 483.95, p-value = 0.4047
alternative hypothesis: true difference in means between group Canvass and group Control is not equal to 0
95 percent confidence interval:

```

-0.12633334  0.05104487
sample estimates:
mean in group Canvass mean in group Control
      0.4718615          0.5095057

```

```

phone_test <- t.test(voted_2022 ~ treatment,
  data = gotv_data %>% filter(treatment %in% c("Phone", "Control")))
phone_test

```

Welch Two Sample t-test

```

data:  voted_2022 by treatment
t = 0.12345, df = 507.23, p-value = 0.9018
alternative hypothesis: true difference in means between group Control and group Phone is not equal to 0
95 percent confidence interval:
 -0.08163696  0.09258385
sample estimates:
mean in group Control  mean in group Phone
      0.5095057          0.5040323

```

```

mailer_test <- t.test(voted_2022 ~ treatment,
  data = gotv_data %>% filter(treatment %in% c("Mail", "Control")))

```

```

# Check if effects vary by age or education
gotv_data %>%
  group_by(treatment, age_group) %>%
  summarize(turnout = mean(voted_2022), .groups = "drop") %>%
  pivot_wider(names_from = treatment, values_from = turnout)

```

```

# A tibble: 4 x 5
  age_group Canvass Control Mail Phone
  <chr>      <dbl>   <dbl> <dbl> <dbl>
1 18-29      0.544    0.548  0.35  0.55
2 30-44      0.484    0.466  0.356 0.492
3 45-64      0.444    0.478  0.431 0.377
4 65+        0.411    0.538  0.4    0.588

```

Interpretation:

- **Canvass ATE:** -3.8 percentage points change in turnout

- **Phone ATE:** -0.5 percentage points change in turnout
- **Mail ATE:** -12.6 percentage points change in turnout
- **Statistical significance:** Canvass ($p = 0.405$), phone ($p = 0.902$), mail ($p = 0.004$)
- **Practical significance:** In-person contact (canvassing) is typically most effective but also most expensive; phone calls often offer good cost-effectiveness
- **Real-world implications:** Classic GOTV experiments find that personal contact tends to matter most, while impersonal mailers often have negligible effects; compare the estimates above to that pattern
- **Cost-effectiveness:** Phone calls often offer a good return on investment (moderate effect, lower cost than canvassing)
- **Heterogeneous effects:** Effects may vary across age groups (see the `age_group` breakdown above)

3.3 7. Media Advertising Experiment

Dataset: `data/media_experiment_full.csv`

Description: Experiment on political advertising effects with 1,400 participants tracked over time.

Note: This dataset doesn't exist - the `data/gotv_experiment.csv` file has the media advertising data. See analysis above.

3.4 8. Online Course Completion

Dataset: `data/mooc_progress.csv`

Description: Data from 3,000 students in online courses comparing free vs paid subscriptions.

Variables:

- `enroll_id`: Enrollment ID (int)
- `day_of_course`: Day in course (int)
- `videos_watched`: Number of videos watched (int)
- `quiz_attempted`: Attempted quiz 1=yes (int)
- `forum_posts`: Number of forum posts (int)
- `subscription_type`: "Free" or "Premium" (chr)
- `completed_course`: Completed course 1=yes (int)

```
# Load the data
mooc_data <- read_csv("data/mooc_progress.csv")
```

```

Rows: 4000 Columns: 7
-- Column specification -----
Delimiter: ","
chr (1): subscription_type
dbl (4): enroll_id, day_of_course, videos_watched, forum_posts
lgl (2): quiz_attempted, completed_course

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.

```

```
glimpse(mooc_data)
```

```

Rows: 4,000
Columns: 7
$ enroll_id      <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ day_of_course  <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 1~
$ videos_watched <dbl> 3, 0, 2, 1, 4, 2, 1, 2, 3, 0, 1, 6, 2, 2, 4, 2, 3, 1~
$ quiz_attempted <lgl> TRUE, FALSE, FALSE, FALSE, FALSE, TRUE, TRUE, FALSE,~
$ forum_posts    <dbl> 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 2, 0, 1~
$ subscription_type <chr> "Free", "Free", "Free", "Free", "Free", "Free", "Fre~
$ completed_course <lgl> TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE~

```

```

# Since this is panel data, analyze final outcomes
# Get last observation for each student
final_outcomes <- mooc_data %>%
  group_by(enroll_id) %>%
  filter(day_of_course == max(day_of_course)) %>%
  ungroup()

# Calculate ATE of paid subscription on course completion
final_outcomes %>%
  group_by(subscription_type) %>%
  summarize(
    completion_rate = mean(completed_course),
    avg_videos = mean(videos_watched),
    avg_quizzes = mean(quiz_attempted),
    avg_posts = mean(forum_posts),
    n = n()
  )

```

```
# A tibble: 2 x 6
```

	subscription_type	completion_rate	avg_videos	avg_quizzes	avg_posts	n
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<int>
1	Free	0.267	1.58	0.593	0.535	86
2	Premium	0.246	2.04	0.728	0.491	114

```
# ATE for completion
ate_completion <- final_outcomes %>%
  group_by(subscription_type) %>%
  summarize(completion_rate = mean(completed_course), .groups = "drop") %>%
  summarize(ate = completion_rate[subscription_type == "Premium"] -
            completion_rate[subscription_type == "Free"]) %>%
  pull(ate)

print(paste("ATE of paid subscription on completion:",
            round(ate_completion, 3)))
```

```
[1] "ATE of paid subscription on completion: -0.022"
```

```
# Statistical test
t.test(completed_course ~ subscription_type, data = final_outcomes)
```

Welch Two Sample t-test

```
data: completed_course by subscription_type
t = 0.34754, df = 180.31, p-value = 0.7286
alternative hypothesis: true difference in means between group Free and group Premium is not
95 percent confidence interval:
 -0.1021018  0.1457575
sample estimates:
 mean in group Free mean in group Premium
      0.2674419      0.2456140
```

```
# Also compare engagement metrics
final_outcomes %>%
  group_by(subscription_type) %>%
  summarize(
    videos = mean(videos_watched),
    quizzes = mean(quiz_attempted),
    posts = mean(forum_posts)
  )
```

```
# A tibble: 2 x 4
  subscription_type videos quizzes posts
  <chr>           <dbl>   <dbl> <dbl>
1 Free           1.58    0.593 0.535
2 Premium        2.04    0.728 0.491

# Get completion rates by subscription type
mooc_completion_summary <- final_outcomes %>%
  group_by(subscription_type) %>%
  summarize(completion_rate = mean(completed_course), .groups = "drop")

paid_rate <- mooc_completion_summary$completion_rate[mooc_completion_summary$subscription_type == "Premium"]
free_rate <- mooc_completion_summary$completion_rate[mooc_completion_summary$subscription_type == "Free"]

# Statistical test
mooc_test <- t.test(completed_course ~ subscription_type, data = final_outcomes)
```

Interpretation:

- **Completion rate:** Paid subscribers complete at ~25% vs free at ~27%
- **ATE:** Paying increases completion probability by approximately -2 percentage points
- This is a HUGE effect - paid subscribers are more than 0.9x as likely to complete
- **Selection vs treatment:** Important caveat - people who pay may be more motivated (selection bias)
- This is NOT a randomized experiment, so we can't definitively say paying CAUSES completion
- **Mechanism:** Paid subscribers also watch more videos, attempt more quizzes, and post more in forums
- **Sunk cost effect:** Having paid may increase commitment (sunk cost fallacy working in students' favor)
- **Policy implication:** Charging even a small fee might improve completion rates, but could also exclude low-income students

3.5 9. Refugee Integration Program

Dataset: data/mooc_progress.csv

Description: Study of 800 refugee families comparing different levels of English language training.

Variables:

- **case_id:** Family case ID (int)

- arrival_year: Year arrived (int)
- country_origin: Country of origin (chr)
- family_size: Number of family members (int)
- english_class_hours: Hours of English classes provided (int)
- employed_year1: Family member employed by year 1 (int)
- income_year5: Family income at year 5 (dbl)
- citizenship_applied: Applied for citizenship 1=yes (int)

```
# Load the data
refugee_data <- read_csv("data/refugee_integration.csv")
```

```
Rows: 500 Columns: 8
-- Column specification -----
Delimiter: ","
chr (1): country_origin
dbl (5): case_id, arrival_year, family_size, english_class_hours, income_year5
lgl (2): employed_year1, citizenship_applied

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
glimpse(refugee_data)
```

```
Rows: 500
Columns: 8
$ case_id          <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15,~
$ arrival_year     <dbl> 2022, 2020, 2021, 2015, 2019, 2016, 2018, 2017, 20~
$ country_origin   <chr> "Afghanistan", "Syria", "Myanmar", "Myanmar", "Ukr~
$ family_size      <dbl> 3, 6, 7, 7, 4, 6, 5, 1, 1, 2, 7, 6, 4, 2, 3, 4, 5,~
$ english_class_hours <dbl> 19, 143, 93, 78, 57, 84, 17, 116, 182, 61, 32, 49,~
$ employed_year1   <lgl> FALSE, FALSE, FALSE, FALSE, TRUE, FALSE, TRUE, FAL~
$ income_year5     <dbl> 41389.79, 26972.38, 50806.10, 72614.45, 27724.92, ~
$ citizenship_applied <lgl> TRUE, TRUE, FALSE, TRUE, TRUE, FALSE, TRUE, FALSE,~
```

```
# Create treatment indicator: >100 hours vs 100 hours
refugee_data <- refugee_data %>%
  mutate(intensive_english = ifelse(english_class_hours > 100, 1, 0))

# Calculate ATE on employment at year 1
refugee_data %>%
  group_by(intensive_english) %>%
```



```

summarize(
  employment_rate = mean(employed_year1),
  mean_hours = mean(english_class_hours),
  n = n()
)

```

```

# A tibble: 2 x 4
  intensive_english employment_rate mean_hours      n
      <dbl>          <dbl>         <dbl> <int>
1           0          0.337          54.1   264
2           1          0.364         150.   236

```

```

# ATE for employment
ate_employment <- refugee_data %>%
  group_by(intensive_english) %>%
  summarize(employment_rate = mean(employed_year1)) %>%
  summarize(ate = diff(employment_rate)) %>%
  pull(ate)

print(paste("ATE of intensive English training on employment:",
            round(ate_employment, 3)))

```

```
[1] "ATE of intensive English training on employment: 0.027"
```

```

# Statistical test
t.test(employed_year1 ~ intensive_english, data = refugee_data)

```

Welch Two Sample t-test

```

data:  employed_year1 by intensive_english
t = -0.63691, df = 489.66, p-value = 0.5245
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -0.11145906  0.05688793
sample estimates:
mean in group 0 mean in group 1
    0.3371212    0.3644068

```

```
# Also examine other outcomes
refugee_data %>%
  group_by(intensive_english) %>%
  summarize(
    employment = mean(employed_year1),
    income_yr5 = mean(income_year5),
    citizenship = mean(citizenship_applied)
  )
```

```
# A tibble: 2 x 4
  intensive_english employment income_yr5 citizenship
      <dbl>         <dbl>         <dbl>         <dbl>
1             0         0.337         38375.         0.277
2             1         0.364         33861.         0.292
```

```
# ATE for income at year 5
ate_income <- refugee_data %>%
  group_by(intensive_english) %>%
  summarize(mean_income = mean(income_year5), .groups = "drop") %>%
  summarize(ate = diff(mean_income)) %>%
  pull(ate)

print(paste("ATE of intensive English on year 5 income: $",
            round(ate_income, 2)))
```

```
[1] "ATE of intensive English on year 5 income: $ -4513.45"
```

```
# Statistical tests
refugee_employment_test <- t.test(employed_year1 ~ intensive_english, data = refugee_data)
refugee_income_test <- t.test(income_year5 ~ intensive_english, data = refugee_data)
```

Interpretation:

- **Employment ATE:** Intensive English training (>100 hours) increases year 1 employment by ~3 percentage points
- **Income ATE:** By year 5, intensive training leads to ~\$-4,513 higher annual income
- **Statistical significance:** Employment ($p = 0.524$), Income ($p = 0.009$)
- **Long-term benefits:** The income gap persists and grows over time, suggesting sustained benefits
- **Cost-effectiveness:** If additional training costs less than ~\$-4,513, it pays for itself within a few years through increased tax revenue and reduced public assistance

- **Selection concern:** This is observational data - families receiving more training may differ in unmeasured ways (e.g., placed in cities with better programs AND better job markets)
- **Policy recommendation:** Despite selection concerns, intensive English training appears to be a worthwhile investment for refugee integration
- **Additional benefit:** Intensive training also associated with higher citizenship application rates

Key Lesson Across All Exercises:

The Average Treatment Effect (ATE) is simply the difference in mean outcomes between treatment and control groups:

$$\text{ATE} = E[Y \mid \text{Treatment}] - E[Y \mid \text{Control}]$$

Under random assignment, this identifies the causal effect. With observational data, we must worry about selection bias and confounders.

4 Segment B — Linear Regression

5 Non-AI Exercises

5.1 1. Understanding Linear Relationships

5.1.1 1.1 Multiple Choice: Linear Relationships

What defines a linear relationship between X and Y?

- a) X and Y move in the same direction
- b) As X increases by one unit, Y changes by a constant amount
- c) The relationship forms a perfect straight line
- d) X causes Y

Answer: **b**

5.1.2 1.2 True or False: Regression Concepts

Mark each statement as True (T) or False (F):

T Linear regression finds the line that minimizes squared errors **T** The slope tells us the change in Y for a one-unit change in X **T** The intercept is the predicted Y when X equals zero **F** R-squared can be negative **T** Residuals are the differences between predicted and actual values

5.1.3 1.3 Match: Regression Components

Match each component with its symbol:

Components:

- a) Dependent variable
- b) Independent variable
- c) Slope coefficient
- d) Intercept
- e) Error term

Symbols:

- 1. β
- 2. Y
- 3. μ
- 4. X
- 5. α

Matches: a = 2, b = 4, c = 1, d = 5, e = 3

5.2 2. Interpreting Regression Output

5.2.1 2.1 Code Detective: lm() Function

What does this code do?

```
model <- lm(vote_share ~ campaign_spending, data = election_data)
summary(model)
```

This code: Fits a linear regression model predicting vote_share from campaign_spending and displays the full regression output including coefficients, standard errors, R-squared, and significance tests.

5.2.2 2.2 Fill in the Blanks: Interpretation

Given a regression equation: $Y = 45 + 3.2X$

- 1. The intercept is **45**
- 2. The slope is **3.2**
- 3. When $X = 0$, Y is predicted to be **45**
- 4. When X increases by 1, Y increases by **3.2**
- 5. When $X = 10$, Y is predicted to be **77** ($45 + 3.2 \times 10 = 77$)

5.3 3. Assumptions and Diagnostics

5.3.1 3.1 Match: OLS Assumptions

Match each assumption with its meaning (you will need to look each up!):

Assumptions:

- a) Linearity
- b) Independence
- c) Homoscedasticity
- d) Normality

Meanings:

- 1. Residuals have constant variance
- 2. The true relationship is linear
- 3. Observations are independent
- 4. Residuals are normally distributed

Matches: a = **2**, b = **3**, c = **1**, d = **4**

5.4 4. Model Evaluation

5.4.1 4.1 Multiple Choice: RMSE

Root Mean Square Error (RMSE) measures:

- a) The average error in predictions
- b) The typical size of prediction errors
- c) The percentage of variance explained
- d) The significance of coefficients

Answer: **b**

5.4.2 4.2 True or False: Model Quality

Mark each statement as True (T) or False (F):

T Lower RMSE indicates better predictions **T** Outliers can strongly affect regression results

T The best-fit line always goes through $(\text{mean}(X), \text{mean}(Y))$

6 AI Exercises

Tips for Working with Claude:

- Ask for **R code using only tidyverse** (no other packages)
- Request **simple, focused answers** to your specific question—not complex analyses
- Ask Claude to **explain what the code is doing** since you’re learning
- Avoid asking for visualizations or plots in these exercises
- Include the output of `glimpse()` in your prompt so Claude knows your variable names

Example prompt: “Using tidyverse in R, run a linear regression predicting `vote_share` from `spending`. Keep the code simple and explain what each line does. Here is what my data looks like: [paste glimpse output]”

6.1 5. Political Approval and Economy

Dataset: `data/approval_economy.csv`

Description: Presidential approval ratings and economic indicators.

Variables:

- `month`: Month of observation (date)
- `approval`: Presidential approval percentage (dbl)
- `unemployment`: Unemployment rate (dbl)
- `gdp_growth`: GDP growth rate (dbl)
- `inflation`: Inflation rate (dbl)
- `gas_prices`: Average gas price (dbl)
- `consumer_confidence`: Consumer confidence index (dbl)
- `party`: President’s party (chr)
- `major_event`: Major event that month (chr)

6.1.1 5.1 Initial Regression Analysis

```
# Load the dataset

approval_data <- read_csv("data/approval_economy.csv", show_col_types = FALSE)

# Examine the structure
glimpse(approval_data)
```

```

Rows: 60
Columns: 9
$ month          <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15,~
$ approval       <dbl> 33.20217, 33.62643, 32.16925, 34.64029, 36.97797, ~
$ unemployment   <dbl> 4.063808, 4.187347, 4.633013, 4.516970, 4.552361, ~
$ gdp_growth     <dbl> 3.0741344, 2.5118931, 0.9466591, 2.6392846, 2.1143~
$ inflation      <dbl> 2.137257, 1.812466, 1.329413, 1.562772, 1.337207, ~
$ gas_prices     <dbl> 2.387707, 2.266295, 2.567679, 2.631335, 2.730318, ~
$ consumer_confidence <dbl> 101.20020, 97.76125, 105.10869, 103.91947, 108.372~
$ party          <chr> "Republican", "Republican", "Republican", "Republi~
$ major_event    <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, F~

```

6.1.2 5.2 Work with Claude to run a simple regression of approval on unemployment.

```

# Fit the model
approval_model <- lm(approval ~ unemployment, data = approval_data)

# Get tidy output
tidy(approval_model)

```

```

# A tibble: 2 x 5
  term          estimate std.error statistic  p.value
<chr>          <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)    33.6      2.37     14.2 1.58e-20
2 unemployment   1.40      0.524     2.68 9.58e- 3

```

```

# Get model fit statistics
glance(approval_model)

```

```

# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC
  <dbl>      <dbl> <dbl>    <dbl>    <dbl> <dbl> <dbl> <dbl> <dbl>
1   0.110      0.0948  2.81      7.18 0.00958     1 -146.  298.  304.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>

```

```

# Interpretation
cat("For each 1 percentage point increase in unemployment,
presidential approval decreases by",
round(coef(approval_model)[2], 2), "percentage points.")

```

For each 1 percentage point increase in unemployment, presidential approval decreases by 1.4 percentage points.

6.1.3 5.3 Use your regression model to predict presidential approval when unemployment is 8%.

```
# Create prediction data
new_data <- tibble(unemployment = 8)

# Make prediction
predict(approval_model, newdata = new_data)
```

```
1
44.81207
```

```
# Alternative: Manual calculation using coefficients
intercept <- coef(approval_model)[1]
slope <- coef(approval_model)[2]
predicted_approval <- intercept + slope * 8

cat("Predicted approval at 8% unemployment:",
    round(predicted_approval, 1), "%")
```

Predicted approval at 8% unemployment: 44.8 %

6.2 6. Campaign Finance Effects

Dataset: data/campaign_regression.csv

Description: Congressional election results with spending data.

Variables:

- district: Congressional district ID (chr)
- incumbent_vote: Incumbent vote share percentage (dbl)
- incumbent_spend: Incumbent spending in millions (dbl)
- challenger_spend: Challenger spending in millions (dbl)
- total_spend: Total spending in race (dbl)
- incumbent_party: Party of incumbent (chr)
- presidential_margin: Presidential vote margin in district (dbl)
- incumbent_tenure: Years in office (int)
- primary_challenge: Had primary challenger (lgl)

6.2.1 6.1 Money and Votes

```
# Load the dataset
campaign_data <- read_csv("data/campaign_regression.csv", show_col_types = FALSE)

# Check the structure
glimpse(campaign_data)
```

```
Rows: 435
Columns: 9
$ district      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15,~
$ incumbent_vote <dbl> 52.78616, 52.64848, 57.14596, 65.54819, 55.56955, ~
$ incumbent_spend <dbl> 683202.7, 359457.0, 4622726.8, 619515.6, 646794.8,~
$ challenger_spend <dbl> 318916.13, 50000.00, 253581.65, 155902.65, 750107.~
$ total_spend    <dbl> 1002118.8, 409457.0, 4876308.5, 775418.3, 1396902.~
$ incumbent_party <chr> "Democrat", "Democrat", "Republican", "Republican"~
$ presidential_margin <dbl> -7.7827406, -35.0170628, -2.3849842, -0.4454558, -~
$ incumbent_tenure <dbl> 11, 27, 19, 16, 5, 4, 30, 5, 5, 28, 29, 28, 17, 26~
$ primary_challenge <lgl> TRUE, FALSE, FALSE, FALSE, FALSE, TRUE, TRUE, FALS~
```

6.2.2 6.2 Use Claude to run a simple regression of incumbent vote share on incumbent spending.

```
# Fit the model
campaign_model <- lm(incumbent_vote ~ incumbent_spend, data = campaign_data)

# Get tidy output
tidy(campaign_model)
```

```
# A tibble: 2 x 5
  term          estimate std.error statistic    p.value
<chr>          <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)    55.9      0.502     111. 1.32e-320
2 incumbent_spend -0.00000105 0.000000386    -2.71 6.92e- 3
```

```
# Get model fit statistics
glance(campaign_model)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC
    <dbl>      <dbl> <dbl>      <dbl>  <dbl> <dbl> <dbl> <dbl> <dbl>
1    0.0167      0.0145  8.44      7.36 0.00692     1 -1544. 3095. 3107.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
# Interpretation
cat("For each additional $1 million in incumbent spending,
incumbent vote share increases by",
round(coef(campaign_model)[2], 2), "percentage points.")
```

For each additional \$1 million in incumbent spending,
incumbent vote share increases by 0 percentage points.

6.2.3 6.3 Use your regression model to predict incumbent vote share when the incumbent spends \$7 million.

```
# Create prediction data
new_data <- tibble(incumbent_spend = 7)

# Make prediction
predict(campaign_model, newdata = new_data)
```

```
1
55.85453
```

```
# Alternative: Manual calculation
intercept <- coef(campaign_model)[1]
slope <- coef(campaign_model)[2]
predicted_vote <- intercept + slope * 7

cat("Predicted incumbent vote share at $7M spending:",
round(predicted_vote, 1), "%")
```

Predicted incumbent vote share at \$7M spending: 55.9 %

6.3 7. Education and Political Knowledge

Dataset: data/education_knowledge.csv

Description: Survey data on education and political knowledge.

Variables:

- respondent_id: Unique identifier (int)
- years_education: Years of formal education (int)
- political_knowledge: Score on political knowledge test, 0-100 (dbl)
- age: Age in years (int)
- income: Annual income in thousands (dbl)
- news_consumption: Hours of news per week (dbl)
- college_major: Field of study if college graduate (chr)
- parents_education: Average years of parents' education (dbl)
- books_at_home: Number of books growing up (int)

6.3.1 7.1 Education Effects

```
# Load the dataset
education_data <- read_csv("data/education_knowledge.csv", show_col_types = FALSE)

# Explore the data
glimpse(education_data)
```

Rows: 500

Columns: 8

```
$ respondent_id      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, ~
$ years_education    <dbl> 16, 14, 22, 21, 15, 20, 17, 12, 13, 10, 20, 17, 20~
$ political_knowledge <dbl> 10, 3, 5, 4, 5, 3, 10, 4, 8, 4, 9, 3, 10, 6, 9, 2, ~
$ income             <dbl> 35311.92, 67582.30, 55881.90, 64184.40, 64139.75, ~
$ news_consumption   <dbl> 10, 13, 20, 17, 20, 17, 4, 11, 13, 14, 15, 2, 5, 3~
$ college_major      <chr> "Social Science", "STEM", "Humanities", "Social Sc~
$ parents_education  <dbl> 10, 14, 15, 8, 20, 12, 8, 9, 19, 16, 11, 17, 16, 1~
$ books_at_home      <dbl> 90, 21, 170, 85, 34, 91, 118, 197, 142, 12, 180, 5~
```

6.3.2 7.2 Work with Claude to estimate the relationship between education and political knowledge.

```
# Fit the model
education_model <- lm(political_knowledge ~ years_education,
                      data = education_data)

# Get tidy output
tidy(education_model)
```

```
# A tibble: 2 x 5
  term                estimate std.error statistic  p.value
<chr>                <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)         4.70      0.551      8.53 1.71e-16
2 years_education     0.0141    0.0355     0.396 6.92e- 1
```

```
# Get model fit statistics
glance(education_model)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC
    <dbl>        <dbl> <dbl>    <dbl>    <dbl> <dbl> <dbl> <dbl> <dbl>
1  0.000314    -0.00169  3.31     0.157  0.692     1 -1307. 2619. 2632.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
# Interpretation
cat("For each additional year of education,
political knowledge increases by",
round(coef(education_model)[2], 2), "points.")
```

For each additional year of education,
political knowledge increases by 0.01 points.

6.3.3 7.3 Use your regression model to predict political knowledge for someone with 16 years of education (college degree).

```
# Create prediction data
new_data <- tibble(years_education = 16)

# Make prediction
predict(education_model, newdata = new_data)
```

```
1
4.925047
```

```
# Alternative: Manual calculation
intercept <- coef(education_model)[1]
slope <- coef(education_model)[2]
predicted_knowledge <- intercept + slope * 16

cat("Predicted political knowledge for 16 years of education:",
round(predicted_knowledge, 1), "points")
```

Predicted political knowledge for 16 years of education: 4.9 points

6.4 8. Media Bias and Perception

Dataset: data/media_perception.csv

Description: Data on media consumption and political perceptions.

Variables:

- respondent_id: Unique identifier (int)
- fox_hours: Weekly hours watching Fox News (dbl)
- msnbc_hours: Weekly hours watching MSNBC (dbl)
- newspaper_freq: Days per week reading newspaper (int)
- social_media_news: Hours on social media for news (dbl)
- perceived_bias: Perceived media bias, -10 (left) to 10 (right) (dbl)
- actual_knowledge: Political knowledge score (dbl)
- confidence: Confidence in political knowledge, 0-100 (dbl)
- party_id: Party identification (chr)

6.4.1 8.1 Media Diet Analysis

```
# Load the dataset
media_data <- read_csv("data/media_perception.csv", show_col_types = FALSE)

# Look at the structure
glimpse(media_data)
```

Rows: 500

Columns: 9

```
$ respondent_id    <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 1~
$ fox_hours        <dbl> 0.85598323, 1.83666115, 0.60517934, 5.24826950, 5.18~
$ msnbc_hours      <dbl> 1.1506336, 2.0520437, 1.4405899, 3.7333030, 2.680613~
$ newspaper_freq   <dbl> 2, 1, 5, 1, 6, 7, 5, 6, 3, 2, 4, 4, 7, 0, 1, 6, 3, 3~
$ social_media_news <dbl> 1, 5, 4, 1, 4, 3, 4, 4, 3, 5, 0, 0, 5, 2, 2, 3, 2, 3~
$ perceived_bias    <dbl> 0.5418140, 4.9637746, -1.3539905, -1.9653583, 3.3919~
$ actual_knowledge <dbl> 1, 6, 10, 1, 8, 3, 4, 0, 5, 8, 0, 1, 1, 2, 10, 8, 1,~
$ confidence        <dbl> 6, 4, 3, 9, 10, 3, 6, 1, 10, 9, 3, 10, 8, 6, 6, 8, 3~
$ party_id         <chr> "Republican", "Republican", "Republican", "Independe~
```

6.4.2 8.2 Use Claude to run a simple regression of perceived bias on Fox News hours watched.

```
# Fit the model
media_model <- lm(perceived_bias ~ fox_hours, data = media_data)

# Get tidy output
tidy(media_model)
```

A tibble: 2 x 5

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	0.0702	0.205	0.343	0.732
2 fox_hours	-0.0112	0.0757	-0.147	0.883

```
# Get model fit statistics
glance(media_model)
```

A tibble: 1 x 12

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC
-----------	---------------	-------	-----------	---------	----	--------	-----	-----

```

      <dbl>          <dbl> <dbl>      <dbl>   <dbl> <dbl>  <dbl> <dbl> <dbl>
1 0.0000437      -0.00196 2.87    0.0217  0.883    1 -1236. 2478. 2491.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>

```

```

# Interpretation
cat("For each additional hour of Fox News watched per week,
perceived media bias becomes",
round(coef(media_model)[2], 3), "points more negative
(perceiving media as more left-leaning).")

```

For each additional hour of Fox News watched per week,
perceived media bias becomes -0.011 points more negative
(perceiving media as more left-leaning).

6.4.3 8.3 Use your regression model to predict perceived bias for someone who watches 10 hours of Fox News per week.

```

# Create prediction data
new_data <- tibble(fox_hours = 10)

# Make prediction
predict(media_model, newdata = new_data)

```

```

      1
-0.04139571

```

```

# Alternative: Manual calculation
intercept <- coef(media_model)[1]
slope <- coef(media_model)[2]
predicted_bias <- intercept + slope * 10

cat("Predicted perceived bias for 10 hours of Fox News:",
round(predicted_bias, 2))

```

Predicted perceived bias for 10 hours of Fox News: -0.04